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# IARCO RESEARCH PROPOSAL

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**Submission Date:** 20th September, 2026

**Research Topic: Climate-related Flood Exposure and Psychosocial Stress as Predictors of Obsessive-Compulsive Symptom Severity Among Adolescents in Constant Flood-prone Regions of Bangladesh: A Cross-Sectional Study.**

# Climate-related Flood Exposure and Psychosocial Stress as Predictors of Obsessive-Compulsive Symptom Severity Among Adolescents in Constant Flood-prone Regions of Bangladesh: A Cross-Sectional Study.

## Introduction:

Climate Change is increasingly getting recognised as a contributing factor in the development of many mental health issues. Many climate-change induced massive disasters like tsunami, wildfires, landslide, river erosion, intense drought and floods being associated with psychological distress and mental health outcomes are working as major reasons behind this [1], [2], [3]. Natural disasters can produce both immediate and long-term psychological consequences at the same time, mostly involving anxiety, depression, post-traumatic stress, and even OCD [1], [2], [4]. Obsessive-compulsive disorder (OCD) is a chronic and impairing condition characterized by significant obsessions and compulsions, with a world-wide prevalence rate of about 2% - 3% [5]. However, literature regarding this issue remains methodologically uneven, with many cross-sectional studies strongly focused in high-income countries rather than climate-vulnerable low and middle income countries [1], [6]. Recent evidence suggests that OCD is caused by both biological vulnerability and environmental distress, including childhood adversity, stressful events, and trauma [7], [8], [9]. Adolescents living in flood affected regions are among those who are significantly exposed to such situations. Although research on climate emotions and eco-anxiety is expanding rapidly, this field yet remains heterogeneous and concentrated in high-income settings, dismissing the need for locally grounded work in disaster-prone countries like Bangladesh [6], [9], [10]. With stigma around mental health still lingering in Bangladesh, vulnerable mental health after disasters are generally overlooked. This study employs a cross-sectional design with random sampling (n=150) of adolescents in a flood-affected region of Bangladesh, collecting Y-BOCS, PSS-10 scores and disaster exposure in a single session and analyzes the data by a two-step hierarchical multiple regression with assumption testing, aiming to investigate whether climate-related flood exposure with psychosocial stress can predict obsessive-compulsive symptom severity among adolescents of flood-affected areas in Bangladesh. This study focuses on revealing and addressing the existing gap in research, connecting OCD and climatic disasters to support early screening in affected communities and to form disaster-responsive sustainable mental health policy framework and proper psychosocial support planning in Bangladesh.

## Research Questions:

1. Is there any positive correlation between recurrence of flood exposure and OCD symptom severity among affected adolescents?

2. Can frequent exposure to climatic disasters act as a precipitant stressor for OCD onset in adolescents already exposed to constant psychosocial stress?

3. Can disaster exposure contribute to additional variance in OCD severity beyond what is already explained by general psychosocial stress?

4. Do adolescents who experienced displacement due to repeated accounts of disasters show OCD severity at a higher rate?

5. How does the stress from frequent disaster exposure impact OCD severity across different age groups (early adolescence 13–15 vs. late adolescence 16–19) or sex?

6. Which specific OCD symptom or type is more noticeable and shows strongest association with climate disaster exposure, compared to others?

**Literature Review:**

According to research, climate change is causing a rise in extreme weather events, and the impacts of these events are not only physical but can also be emotional [1], [2], [3]. There has been an increase in the number of people who survived severe flooding, which increases the risk of having mental disorders like depression, anxiety, PTSD, and other related problems [1], [3]. The influence of such an event on an individual can be so severe that it can even trigger obsessive-compulsive disorder [5]. Adolescents are particularly vulnerable to the effects of such weather-related disasters. In their formative years, teenagers are excessively reliant on the security and stability of their homes, schools, and social groups [8]. The destruction of one's home environment and forced migration generate significant psychological distress in adolescents. Such distress, according to recent studies, is determined by one's sex and age: the higher the socioeconomic status and cognitive development, the higher is the likelihood of developing eco-anxiety [6], [9], [10], whereas female adolescents tend to report higher incidence of OCD and PTSD following weather-related disasters [7]. For teenagers, successive exposure to such man-made and natural disasters acts as a constant stress-inducing factor, severely impacting the already unstable mental health of adolescents, with higher rates of anxiety and OCD as a result [5], [7]. Though most of the studies cited illustrate the effect of climate change on weather patterns and the effect on the environment, infrastructure, and the economy, few relate directly to psychological conditions in adolescents. The tendency, however, seems apparent: as the impact of climate change becomes impossible to ignore, both teenagers and adults experience mental health issues related to the changing environment [1], [2]. In most cases, such disasters are accentuated for low-income countries, where the response to climate change is severely limited by political, social, and economic factors. Bangladesh, for example, is a lower-middle-income country with limited resources to adapt to the effects of climate change. River bank erosion in Bangladesh, which tends to occur frequently in rural regions, not only destroys infrastructure but entire households' livelihoods [4]. As a result, the incidence of depression and anxiety in adolescents is particularly acute. Apart from a lack of resources, Bangladesh has cultural and social barriers to mental health care, meaning rural adolescents have little to no help in dealing with the disaster [4]. For such communities, a weather-related disaster is a catalyst for a severe mental health episode. The type of disaster seems to correlate with the type of mental health issue following it. Flooding, for example, often results in water-borne illness; the pervasive and frequent threat to general health resulting from such weather-related disasters often transforms into delusions of germ infestation with OCD symptoms as a result [5]. Additionally, the threat of property loss during a flood is much higher, and the psychological response to such a catastrophe is sheer helplessness, which manifests itself in the desire to check and recheck that everything is in order [5]. Amongst these extensive studies only few have explored the connection between disaster exposure and Obsessive Compulsive Symptoms, keeping this aspect vague without clear analysis and investigation. Hence, further research should be conducted to better understand the relationship between adolescents' mental health and exposure to weather-related disasters like flood. This study aims to conduct extensive analysis and investigation to identify this

relationship to initiate adolescence directed disaster-focused mental health policy and support systems to care for vulnerables in affected areas, to form better climate policies and contribute towards local in-depth research on this issue.

**Methodology:**

**Study Design:** This study will adopt an analytical cross-sectional design to investigate if frequent flood exposure and psychosocial stress can predict OCD symptoms among adolescents of flood-affected regions in Bangladesh. Cross-sectional designs are generally used for climate-mental health research, though there are some limitations in establishing causal or temporal relationships [1], [11]. Given the large gap in prior research, this design is appropriate as a perfect preliminary step, connecting flood exposure to OCD specifically.

**Sample and Sample Size:** A random sample of adolescents (n=150) of age between 13 to 19 will be recruited from a flood-prone region of Bangladesh, using records of recurrent flood history, consistent with the methodologies used in previous climate-mental health studies of Bangladesh [12], [13].

**Inclusion and Exclusion Criteria:**

**Inclusion:** teenagers aged between 13 to 19; inhabitant in the selected flood-affected area for at least 12 months; exposure to at least one flood; understands Bangla; adolescent assent with parental consent. **Exclusion:** teenagers with severe cognitive or communication difficulties and acutely sick with medical and psychiatric emergencies, but having OCD symptoms, anxiety and depression won't exclude participants.

**Data Collection Method:** Data will be collected in one 30 to 45 minute session for each participant. This session will include consent or assent, a demographic questionnaire, a flood-exposure and displacement questionnaire, the PSS-10, and the Y-BOCS (adolescent-adapted). Trained assessors will give these assessments.

**Instruments:** The primary outcome measure for OCD symptom severity is Y-BOCS (Yale-Brown Obsessive Compulsive Scale); stress is well documented as a factor that influences the expression and onset of OCD symptoms [5], [12], [14].

**Flood Exposure Questionnaire (specific to the study):** A composite score based on flood frequency, displacement, and loss, derived from flood exposure measurement methods used in Bangladesh [11], [15].

**Displacement Item:** It will be measured separately, since displacement has a separate psychosocial pathway as outlined in the Bangladesh climate-immobility research [13].

**PSS-10:** It measures general psychosocial stress. Hence, it will address the aim of this study concerning isolation of the flood exposure's independent contribution.

**Types of Variables:** Dependent: OCD symptom severity (continuous, Y-BOCS). Independent: flood exposure score (continuous), displacement status (categorical). Control: PSS-10 (continuous). Covariates: age, sex, socioeconomic status, prior mental-health history.

**Descriptive Statistics:** Means/SDs (or medians/IQR) for continuous variables; percentages for categorical variables; Cronbach's alpha for PSS-10 and Y-BOCS.

**Inferential Statistics:** Bivariate correlations will be addressing RQ1. A two-step hierarchical regression (Model 1: demographics + PSS-10; Model 2: flood exposure and displacement) will test whether exposure explains additional variation by $\Delta R^2$ (RQ2, RQ3). T-tests will be comparing displacement groups (RQ4). Exploratory interaction terms (exposure × age group; × sex) will examine the differential effects (RQ5), and where available, dimensional analysis with false-discovery-rate correction will determine the OCD dimension mostly linked with exposure (RQ6), aligned with previous stress-vulnerability research [7], [8], [14]. Assumptions (normality, multicollinearity, homoscedasticity) will be tested; $\alpha$=0.05, with 95% CIs reported.

SPSS or R softwares will be used for data analysis**.** This cross-sectional design cannot establish causation or onset timing; exposure recall may result in recall bias; n=150 may be small for subgroup analyses; and the results may not generalize beyond the study region [1], [4].

**Expected Outcomes:**

23 The study will show a positive correlation between the amount of flood exposure and the severity of OCD symptoms among adolescents of the study area, aligned with the broader evidence, connecting climate-related stressors to adverse mental health damages [1], [2], [16]. It is anticipated that flood exposure will account for some of the variation in OCD severity that cannot be explained by general psychosocial stress alone, thus supporting the idea that climate-related stress acts as a separate contributing factor rather than just overlapping with other known stressors [5], [14]. Adolescents who have been displaced are expected to display relatively greater OCD symptom severity, mirroring the distinct psychological burden of displacement that has been documented in climate research conducted in Bangladesh [12], [15]. The study will also show differences in vulnerability according to age or sex, and it might identify which particular dimensions of OCD symptoms are most closely linked to disaster exposure, providing a more detailed understanding [7], [8]. Since the sample used in this study is small and exploratory, the results should be seen as preliminary associations that need to be confirmed through further research involving larger samples rather than as established causal relationships.

**Conclusion:**

This research aims to contribute in filling the critical evidence gap in an area of intersection 29 between climate change and adolescence mental health in Bangladesh. Many of these young people specifically, living in lower or coastal areas, are highly vulnerable to recurring flood disasters and climate-related displacements [11], [13]. The literature on climate related distress 32 and its association with mental health outcomes, including anxiety, depression, and other common mood disorders, is growing steadily [1], [2], [3]. In contrast, limited attention has been paid to the connection between disaster exposure and OCD symptom severity. This study will attempt to empirically establish this association for the first time, hence contributing to the evidence base for early intervention and support for risky populations. If this study finds the predicted associations, then it will provide a foundation for future longitudinal research that would be able to determine the direction of the relationships and identify potential causal mechanisms, contributing to the development of more targeted, evidence-based intervention strategies. In addition, this will help shape climate-sensitive mental health policy and planning for Bangladesh and similar climate-exposed countries and regions.

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